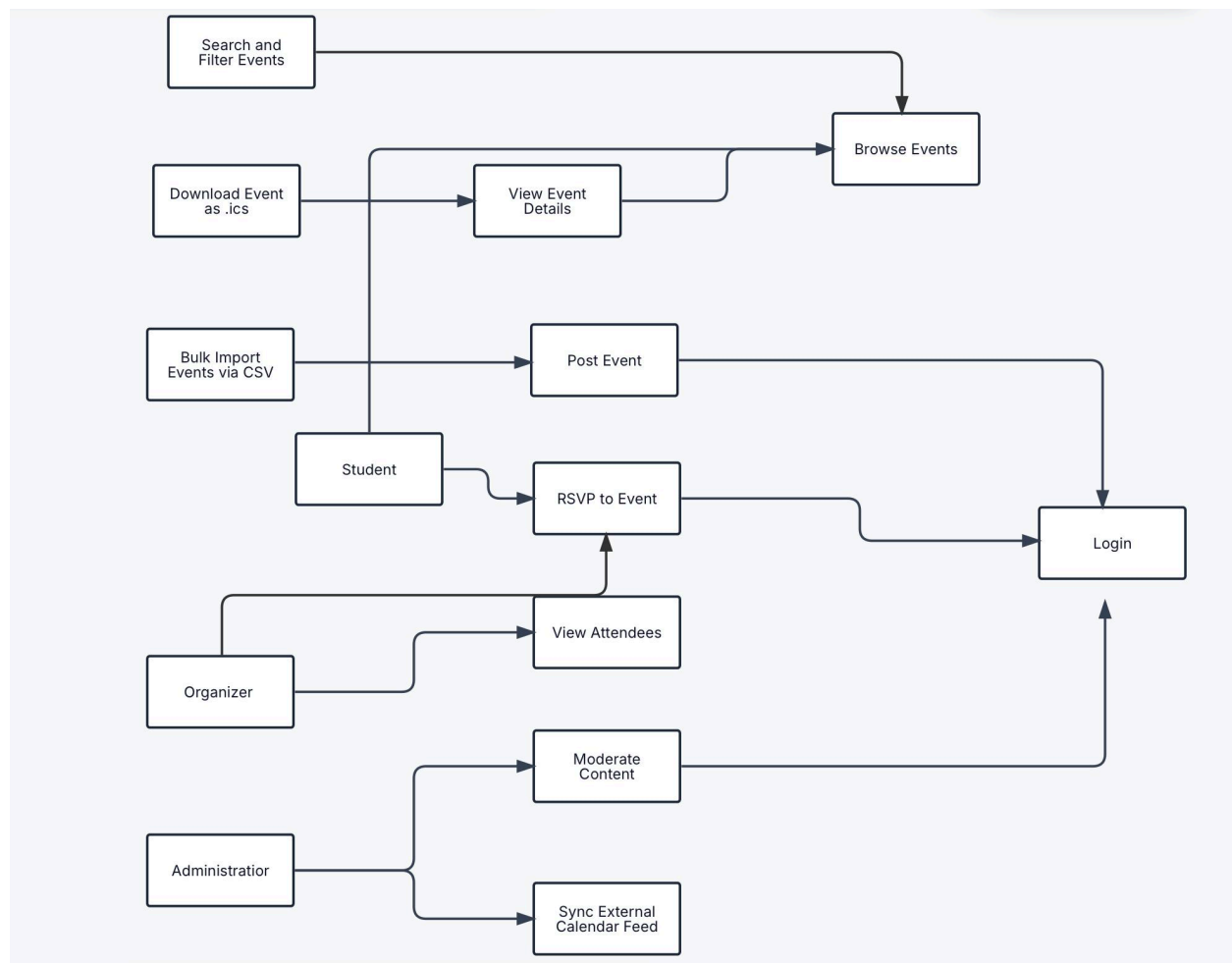


Campus Event Board: UML Diagrams

1. Use Case Diagram

The system boundary is Campus Event Board. The actors are Student, Organizer, and Administrator. An organizer is also able to do anything a student can do (browse, search, RSVP), and an administrator is also able to do anything an organizer can do, but the diagram below shows only each actor's primary use cases to keep it readable.



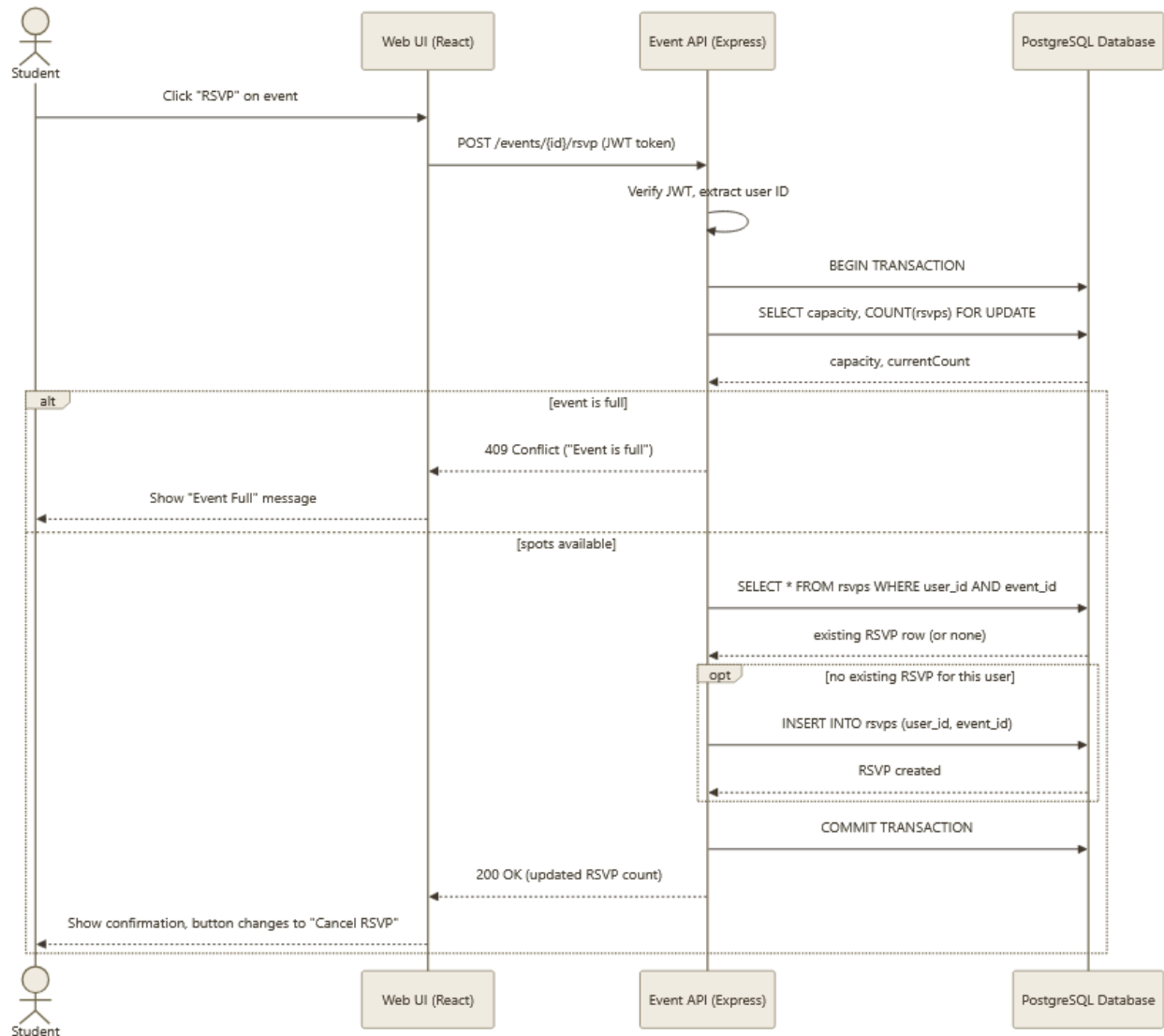
Use Case Descriptions

Use Case	Primary Actor	Description
----------	---------------	-------------

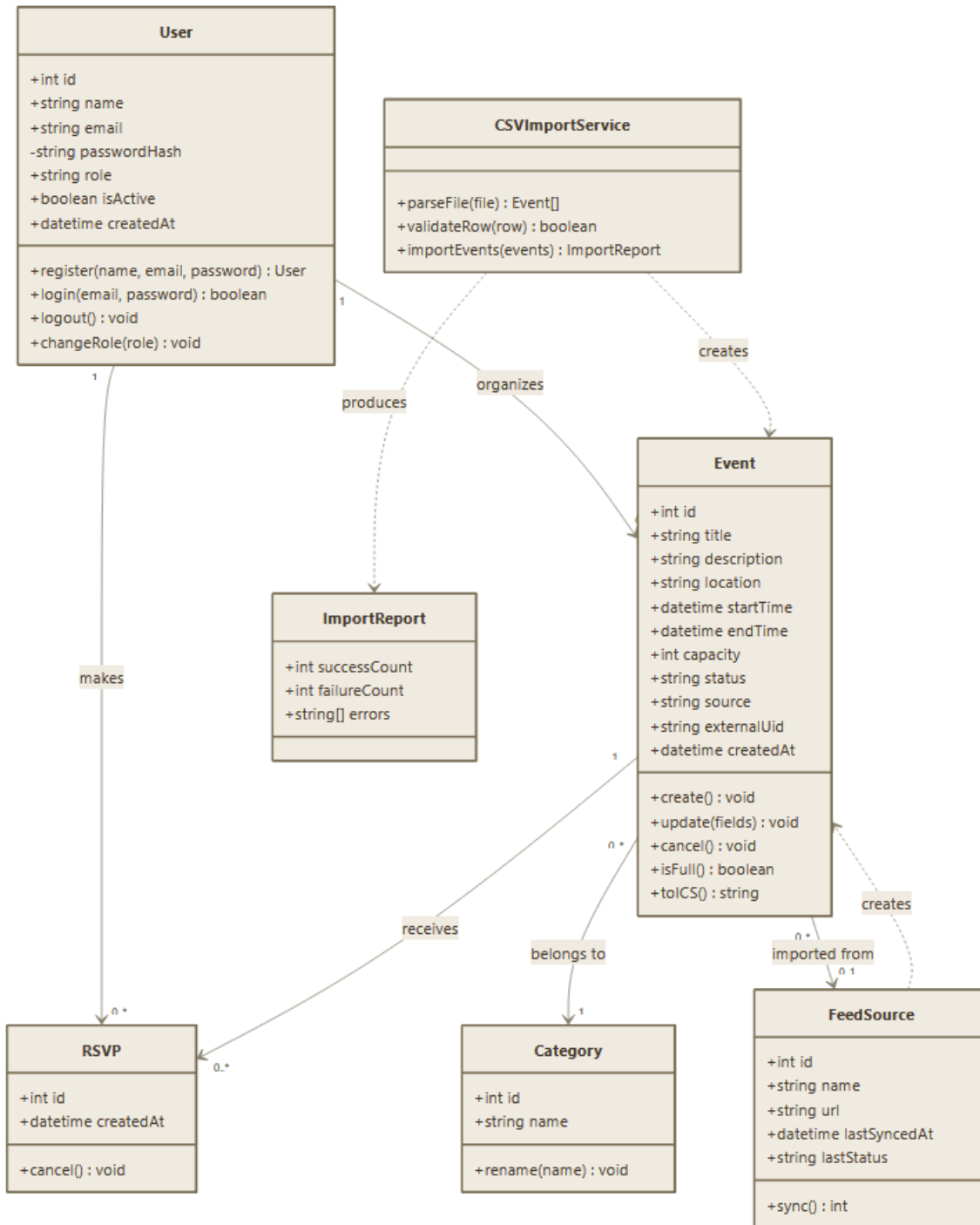
Browse Events	Student	The actor views the list of upcoming events, sorted by start time, without needing to log in. This is the base use case for discovering events.
Search / Filter Events <i>(extends Browse Events)</i>	Student	The actor narrows the event list by keyword, category, date range, or location. This is an optional refinement of browsing, so it extends rather than always executes.
View Event Details <i>(extends Browse Events)</i>	Student	The actor opens a single event to see its full description, location, time, organizer, and current RSVP count.
Login	Student, Organizer, Administrator	The actor authenticates with an email and password. This use case is included by any action that requires knowing who the user is.
RSVP to Event <i>(includes Login)</i>	Student	A logged-in actor reserves a spot at an event, or cancels an existing reservation. Login is always required first, so it is included rather than optional.
Download Event as .ics <i>(extends View Event Details)</i>	Student	From the event detail page, the actor optionally downloads the event as a calendar file to import into their own calendar app.
Post Event <i>(includes Login)</i>	Organizer	A logged-in organizer creates a new event by filling out a form with title, description, category, location, time, and optional capacity.
Bulk Import Events via CSV <i>(extends Post Event)</i>	Organizer	Instead of creating one event at a time, the organizer uploads a CSV file of many events as an alternate flow of posting events.
View Attendees	Organizer	The organizer views the list of students who have RSVP'd to one of their own events.
Moderate Content <i>(includes Login)</i>	Administrator	A logged-in administrator removes an event, deactivates a user account, or manages the category list.
Sync External Calendar Feed	Administrator	The administrator registers a public ICS feed URL; the system imports new events from it and skips events that already exist.

2. Sequence Diagram: RSVP to Event

This is the application's central use case: it touches the UI, the authenticated API, and the database, and it must behave correctly even when two students RSVP at the same moment, which makes it a good representative of the whole system's behavior. The diagram includes an alt fragment (event full vs. spots available) and a nested opt fragment (skip if already RSVP'd).



3. Class Diagram



Relationship notes:

- A User (as organizer) organizes 0 or many Events; each Event has exactly 1 organizing User.

- A User makes 0 or many RSVPs; each RSVP belongs to exactly 1 User. Likewise, an Event receives 0 or many RSVPs, and each RSVP belongs to exactly 1 Event — so RSVP is the association (join) class between User and Event, enforcing one RSVP per user per event.
- Each Event belongs to exactly 1 Category, and a Category can have 0 or many Events.
- Each Event is imported from 0 or 1 FeedSource (null if posted manually or via CSV); a FeedSource can produce 0 or many Events.
- CSVImportService and FeedSource have dependency relationships with Event, since they create Event instances but do not own them long-term the way User does.